

My thinking on this is Beryllium is the 4th element
 and if we start from ground state $n=0$, then with
 Beryllium having four electrons ($n=3$), we combine states
 which would be $\frac{0}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \frac{3}{2}$.

I could be wrong, and if I am in fact, incorrect
 then it's unfortunate how this lines up...

5) Difference between $|\uparrow\rangle$ and the $|1\rangle$:

$\Rightarrow |\uparrow\rangle$ indicates $\frac{1}{2}$ spin or "spin- $\frac{1}{2}$ states" (C. Monroe, B. P. King -
 in this case spin-up Physical Review Letters

$\Rightarrow |1\rangle$ indicates second quantized state for harm. osc.

18/12/1995

6) Carrier transitions and red/blue sideband transitions:

$\Rightarrow$ Applying a pair of off-resonant laser beams to ion
 (Raman transitions). When freq diff. of beams $(\delta) \approx \omega_0$,
 which is known as carrier, coherence between states, $\delta \approx \omega_0 - \omega_x$
 or the red sideband, $\delta \approx \omega_0 + \omega_x$ or blue sideband (transitions between
 (transitions between $|0\rangle|\downarrow\rangle$ and $|1\rangle|\uparrow\rangle$) $|1\rangle|\downarrow\rangle$ and $|0\rangle|\uparrow\rangle$)

7) $|aux\rangle$ state: auxiliary atomic level

$\Rightarrow$ this "simply reverses the sign of any component
 of the register in the $|1\rangle|\uparrow\rangle$ " by
 gifting a Rabi cycle

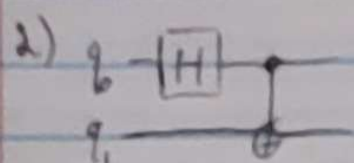
$\Rightarrow |1\rangle|\uparrow\rangle \rightarrow |0\rangle|aux\rangle \rightarrow -|1\rangle|\uparrow\rangle$

9) spring const.: harmonic osc. $\therefore$ x-direction

Ryan Saelens

1) 4x4 Matrix CNOT Gate

$$\Rightarrow \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}$$



* This circuit should yield entangled bell state $|\Phi\rangle$

$$\begin{aligned} \begin{matrix} |0\rangle \\ |1\rangle \end{matrix} & \begin{matrix} \left(\frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle \right) |1\rangle \\ \end{matrix} \\ & = \frac{1}{\sqrt{2}} |01\rangle + \frac{1}{\sqrt{2}} |11\rangle \xrightarrow{\text{CNOT}} \frac{1}{\sqrt{2}} |01\rangle + \frac{1}{\sqrt{2}} |10\rangle \\ & \quad \downarrow \\ & = |\Phi\rangle \end{aligned}$$

* Thoughts: If I recall correctly, the above state, is also known as $|\Psi^+\rangle$.

3) Meanings for "2", "5" and " $\frac{1}{2}$ " in $^2S_{\frac{1}{2}}$ for $^9\text{Be}^+$

- $\Rightarrow$ 2 indicates total number of spin states
- $\Rightarrow$ 5 indicates orbital angular momentum
- $\Rightarrow$ $\frac{1}{2}$ combined angular momentum or nuclear spin

4) Two nuclear hyperfine levels are $|F=2, m_F=2\rangle$ and $|F=1, m_F=1\rangle$, why then does nuclear spin for ^9Be have to be $\frac{3}{2}$?

* We know that if there are more than one electron in orbital they combine to give values over $\frac{1}{2}$